Ps 05

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1 Problem 2.1

1.1 Task

(Parallelogram law & polarization identity)

Let $\mathbb{E} = \mathbb{R}^n$. Prove that

$$\|\cdot\|_1 : \mathbb{E} \to \mathbb{R}^+$$

given by

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|, \quad x = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$$

is a norm on $\mathbb{R}^n$, but is not induced by an inner product for $n \geq 2$.

1.2 Solution

1. $\|\cdot\|_1$ is a norm

   Let $x, y \in \mathbb{R}^n, \lambda \in \mathbb{R}$.

   - non-negativity
     $|x_i| \geq 0 \implies \|x\|_1 = \sum_{i=1}^{n} |x_i| \geq 0$.
   - non-degeneracy
     If $\|x\|_1 = 0$, then $\sum_{i=1}^{n} |x_i| = 0$ with non-neg summands, so $|x_i| = 0 \, \forall i \iff x = 0$.
   - homogeneity

     $$\|\lambda x\|_1 = \sum_{i=1}^{n} |\lambda x_i| = |\lambda| \sum_{i=1}^{n} |x_i| = |\lambda| \|x\|_1$$

   - triangle inequality

     $$\|x + y\|_1 = \sum_{i=1}^{n} |x_i + y_i| \leq \sum_{i=1}^{n} (|x_i| + |y_i|) = \|x\|_1 + \|y\|_1$$

2. Not induced by any inner product for $n \geq 2$

   Theorem 2.8: Jordan–von Neumann. A norm $\|\cdot\|$ is induced by an inner product iff it satisfies the parallelogram identity:

   $$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2), \quad \forall x, y \in \mathbb{E}$$

   By contraposition, one counterexample pair suffices. Take $x = e_1, y = e_2$ (pad with zeros for $n > 2$):

   - $\|x\|_1 = \|y\|_1 = 1$
   - $x + y = (1, 1, 0, \ldots, 0) \implies \|x + y\|_1 = 2$

- $x - y = (1, -1, 0, \ldots, 0) \implies \|x - y\|_1 = 2$

$$\|x + y\|_1^2 + \|x - y\|_1^2 = 4 + 4 = 8 \neq 4 = 2(1 + 1) = 2(\|x\|_1^2 + \|y\|_1^2)$$

Parallelogram identity fails $\implies \|\cdot\|_1$ is not induced by any inner product. ■

2 Problem 2.4

2.1 Task

Let $E = \mathbb{R}^3$, and consider the vectors $x, y, z \in \mathbb{R}^3$ expressed in coordinates from the canonical basis $\mathcal{E} = \{e_1, e_2, e_3\}$:

$$\begin{aligned} [x]^{\mathcal{E}} &= (1, 0, 1)^T \\ [y]^{\mathcal{E}} &= (0, 3, 1)^T \\ [z]^{\mathcal{E}} &= (-5, 1, 3)^T \end{aligned}$$

Show that these vectors form a basis of $\mathbb{R}^3$, and compute the orthonormal basis obtained by their Gram-Schmidt orthonormalization.

2.2 Solution

1. $\{x, y, z\}$ is a basis of $\mathbb{R}^3$

   Three vectors in $\mathbb{R}^3$ form a basis iff they are linearly independent (dimension argument: $\dim \mathbb{R}^3 = 3$, so independence $\implies$ spanning).

   - Set up $\alpha x + \beta y + \gamma z = 0$ componentwise:

   $$\begin{cases} \alpha - 5\gamma = 0 \\ 3\beta + \gamma = 0 \\ \alpha + \beta + 3\gamma = 0 \end{cases}$$

   - Row 1: $\alpha = 5\gamma$. Row 2: $\beta = -\frac{\gamma}{3}$. Substitute into row 3:

   $$5\gamma - \tfrac{\gamma}{3} + 3\gamma = \tfrac{23\gamma}{3} = 0 \implies \gamma = 0 \implies \alpha = \beta = 0$$

   - Only the trivial solution $\implies$ linearly independent $\implies$ basis.

2. Gram–Schmidt orthonormalization

   Set $x_1 := x, x_2 := y, x_3 := z$. Standard inner product $\langle a, b\rangle = a^T b$. Recursion (Thm 2.13):

   $$e_1 := \frac{x_1}{\|x_1\|}, \qquad e_i := \frac{x_i - \sum_{j=1}^{i-1} \langle x_i, e_j\rangle e_j}{\|x_i - \sum_{j=1}^{i-1} \langle x_i, e_j\rangle e_j\|}, \quad 2 \leq i \leq n$$

- Step 1
  $\|x_1\|^2 = 1 + 0 + 1 = 2 \implies \|x_1\| = \sqrt{2}$

$$e_1 = \tfrac{1}{\sqrt{2}}(1, 0, 1)^T$$

- Step 2
  $\langle x_2, e_1\rangle = \frac{1}{\sqrt{2}}(0 + 0 + 1) = \frac{1}{\sqrt{2}}$

$$\langle x_2, e_1\rangle e_1 = \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}}(1, 0, 1)^T = \tfrac{1}{2}(1, 0, 1)^T$$

$$x_2 - \langle x_2, e_1\rangle e_1 = (0, 3, 1)^T - \tfrac{1}{2}(1, 0, 1)^T = (-\tfrac{1}{2}, 3, \tfrac{1}{2})^T$$

  $\|\cdot\|^2 = \frac{1}{4} + 9 + \frac{1}{4} = \frac{19}{2} \implies \|\cdot\| = \sqrt{\frac{19}{2}}$
  Normalize (multiply num. and denom. by 2 to clear fractions): $(-\frac{1}{2}, 3, \frac{1}{2})^T \cdot 2 = (-1, 6, 1)^T$, and $\sqrt{19/2} \cdot 2 = \sqrt{38}$.

$$e_2 = \tfrac{1}{\sqrt{38}}(-1, 6, 1)^T$$

- Step 3
  $\langle x_3, e_1\rangle = \frac{1}{\sqrt{2}}(-5 + 0 + 3) = \frac{-2}{\sqrt{2}} = -\sqrt{2}$
  $\langle x_3, e_2\rangle = \frac{1}{\sqrt{38}}(5 + 6 + 3) = \frac{14}{\sqrt{38}}$

$$\langle x_3, e_1\rangle e_1 = -\sqrt{2} \cdot \tfrac{1}{\sqrt{2}}(1, 0, 1)^T = -(1, 0, 1)^T$$

$$\langle x_3, e_2\rangle e_2 = \tfrac{14}{\sqrt{38}} \cdot \tfrac{1}{\sqrt{38}}(-1, 6, 1)^T = \tfrac{14}{38}(-1, 6, 1)^T = \tfrac{7}{19}(-1, 6, 1)^T$$

$$x_3 - \langle x_3, e_1\rangle e_1 - \langle x_3, e_2\rangle e_2 = (-5, 1, 3)^T + (1, 0, 1)^T - \tfrac{7}{19}(-1, 6, 1)^T$$

$$= (-4, 1, 4)^T - \tfrac{1}{19}(-7, 42, 7)^T = \tfrac{1}{19}\big((-76, 19, 76)^T - (-7, 42, 7)^T\big) = \tfrac{1}{19}(-69, -23, 69)^T = \tfrac{23}{19}(-3, -1, 3)^T$$

  $\|\cdot\| = \frac{23}{19}\sqrt{9 + 1 + 9} = \frac{23}{19}\sqrt{19} = \frac{23}{\sqrt{19}}$

$$e_3 = \frac{\frac{23}{19}(-3,-1,3)^T}{\frac{23}{\sqrt{19}}} = \tfrac{1}{\sqrt{19}}(-3, -1, 3)^T$$

- Verification
  $\propto$ means "equal up to a positive scalar"; the normalizing factors $\frac{1}{\sqrt{2}\sqrt{38}}$ etc. cannot affect whether the inner product is zero, so only the integer direction vectors matter.
  $\langle e_1, e_2\rangle \propto (1, 0, 1) \cdot (-1, 6, 1) = -1 + 0 + 1 = 0$
  $\langle e_1, e_3\rangle \propto (1, 0, 1) \cdot (-3, -1, 3) = -3 + 0 + 3 = 0$
  $\langle e_2, e_3\rangle \propto (-1, 6, 1) \cdot (-3, -1, 3) = 3 - 6 + 3 = 0$ ■

3 Problem 2.7

3.1 Task

(Inner product and positive definiteness)

In $\mathbb{R}^2$ we define $\omega(x, y) := xSy$, where

$$S = \begin{pmatrix} 1 & p \\ p & q \end{pmatrix}$$

Prove that $\omega : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ is an inner product if and only if $\Delta = q - p^2 > 0$. For which values of (p, q) is the standard basis an orthonormal basis w.r.t. ω? Can ω be an inner product if $p < 0$? What about when $q \leq 0$?

3.2 Solution

$\omega(x, y) = x^T Sy$ with S symmetric ($S^T = S$), so ω is automatically bilinear and symmetric. Only positive-definiteness is at stake. Setting $y = x$:

$$\omega(x, x) = x_1^2 + 2p\, x_1 x_2 + q\, x_2^2$$

1. ω is an inner product $\iff \Delta = q - p^2 > 0$

   Complete the square in x_1:

   $$\omega(x, x) = (x_1 + px_2)^2 + (q - p^2)\, x_2^2$$

   - ($\Leftarrow$) Assume $\Delta > 0$. Both summands are ≥ 0 and vanish together only when $x_2 = 0$ and $x_1 + px_2 = x_1 = 0$, i.e., $x = 0$. So $\omega(x, x) > 0$ for $x \neq 0$.
   - ($\Rightarrow$) Contrapositive. If $\Delta \leq 0$, take $x = (-p, 1)^T \neq 0$. The first square vanishes, giving $\omega(x, x) = \Delta \leq 0$, contradicting positive-definiteness.

2. Standard basis orthonormal w.r.t. ω

   In general $\omega(e_i, e_j) = S_{ij}$, so $\{e_1, e_2\}$ orthonormal w.r.t. $\omega \iff S = I$:

   - $\omega(e_1, e_1) = 1$ (always).
   - $\omega(e_2, e_2) = q \overset{!}{=} 1$.
   - $\omega(e_1, e_2) = p \overset{!}{=} 0$.

   So $(p, q) = (0, 1)$, and ω reduces to the standard dot product. Consistency: $\Delta = 1 > 0$.

3. Can $p < 0$?

   Yes. The condition $q > p^2$ depends only on $|p|$, not on the sign of p. Example: $(p, q) = (-1, 2) \implies \Delta = 1 > 0$.

4. Can $q \leq 0$?

   No. $\omega(e_2, e_2) = q$, and positive-definiteness requires $\omega(e_2, e_2) > 0$, hence $q > 0$. Equivalently, $q > p^2 \geq 0 \implies q > 0$.

   Asymmetry worth noting: p is unconstrained in sign (off-diagonal cross-term), but q must be strictly positive (diagonal entry = squared \$$\omega$\$-norm of e_2). ■

4 Problem 2.8

4.1 Task

(Complex field as matrices)

Given $\mathbb{E} = \mathbb{R}^{2\times 2}$ the space of real two-by-two matrices, we consider the subspace $\mathbb{F} = \text{span}(\text{Id}, J)$, where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

We define the following bilinear form:

$$\omega(A, B) = \frac{1}{2}\text{Tr}(B^T A)$$

where $\text{Tr}(A)$ is the trace of a matrix A.

(i) Prove that ω is an inner product on $\mathbb{F}$, and that $\{\text{Id}, J\}$ forms an orthonormal basis w.r.t. ω.

(ii) We define the following linear function $\Phi : \mathbb{C} \to \mathbb{F}$ given by:

$$\Phi(z) = \Re(z) \cdot \text{Id} + \Im(z) \cdot J = \begin{pmatrix} \Re(z) & -\Im(z) \\ \Im(z) & \Re(z) \end{pmatrix} \in \mathbb{F}, \quad \forall z \in \mathbb{C}$$

where $\Re(z)$ and $\Im(z)$ are the real and imaginary part of z. Prove that Φ is an isomorphism between $\mathbb{C}$ and $\mathbb{F}$, and that for any complex numbers z_1, z_2 we have

$$\Phi(z_1 \cdot z_2) = \Phi(z_2) \cdot \Phi(z_1)$$

(iii) Let $\|\cdot\| : \mathbb{F} \to \mathbb{R}^+$ be the norm associated to ω, namely $\|\cdot\| := \sqrt{\omega(\cdot, \cdot)}$. Given a complex number $z \in \mathbb{C}$, what does $\|\Phi(z)\|$ represent? (**Indication:** One may first consider this: what does the $\Phi(z)^T$ represent?)

4.2 Solution